

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459336.7906 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.021
Transit depth (Rp/Rs)^2	2.13 +/- 0.6 [%]
Orbital Inclination (inc)	85.2 +/- 0.92
Airmass coefficient 1 (a1)	1.123 +/- 0.012
Airmass coefficient 2 (a2)	-0.106 +/- 0.04
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#3 - [484, 210]
Optimal Aperture	3.51
Optimal Annulus	11.11
Transit Duration (day)	0.062 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.146 ± 0.021 vs 0.1063 (deviation 1.89σ)
Duration fit vs published	0.062 d vs 0.0867 d (deviation 1.17σ)
Mid-time O–C	-9.5 min
Depth SNR	7.0
Residual scatter	1.08 %

Light curve

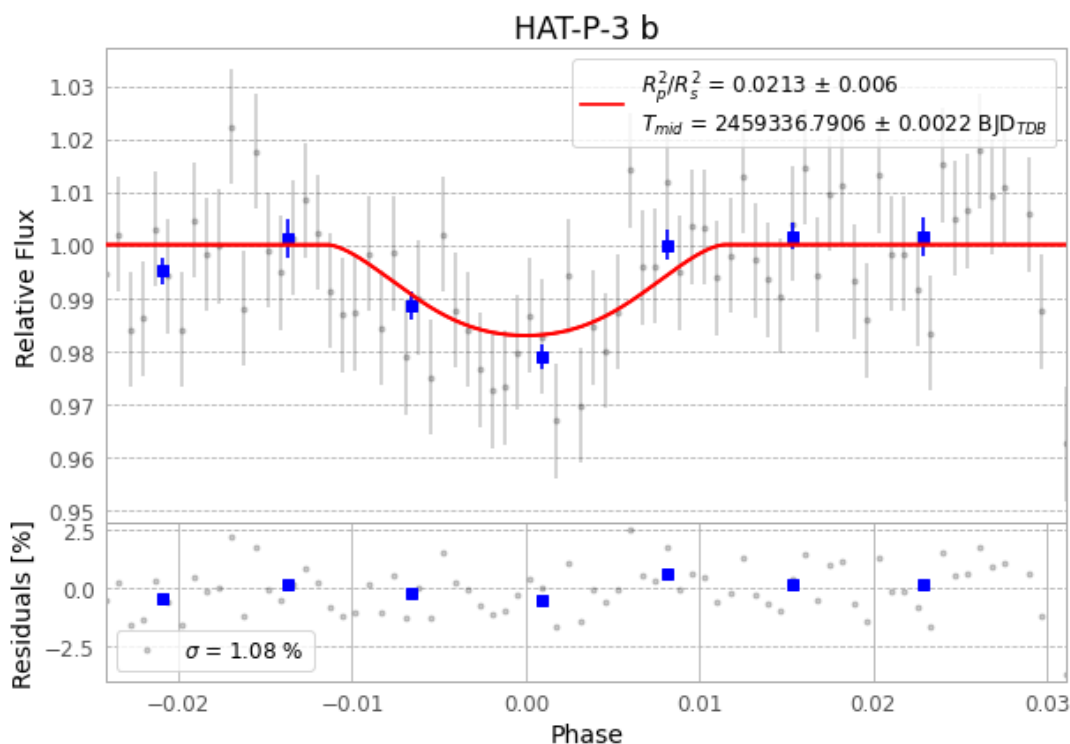


Figure 1 — FinalLightCurve_HAT-P-3 b_2021-05-01.png (SHA-256 48bac05b40d9ea0b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 447677c51a2ff2eb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), HATP-3210502051212.FITS → HATP-3210502090612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-210502.log (5937c334cade...), FinalLightCurve_HAT-P-3 b_2021-05-01.png (48bac05b40d9...), FinalLightCurve_HAT-P-3 b_2021-05-01.csv (ca3e10b7ceed...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 09:08Z.